

ON THE RIEMANN-HURWITZ THEOREM AND ITS APPLICATIONS

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ABSTRACT. This work is an exposition of the Riemann-Hurwitz theorem and some of its applications, such as genus computation, the Mason-Stothers theorem (which is related to the abc conjecture) and Hurwitz's automorphisms theorem. The proof of the Riemann-Hurwitz formula that we present is topological. Thus, this work can serve as a motivation for the study of topological methods in algebraic geometry.

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1. PRELIMINARIES

A Riemann surface is a topological space X which is Hausdorff, second countable, connected, and has a complex structure $\mathcal{A}$ of dimension 1 over $\mathbb{C}$. Every smooth projective curve in $\mathbb{P}^2$ is a compact Riemann surface of genus $(n-1)(n-2)/2$. A map $F: X \rightarrow Y$ between Riemann surfaces is *holomorphic* (resp. *meromorphic*) at $p \in X$ if, for every chart $\phi_1: U_1 \subseteq X \rightarrow V_1 \subseteq \mathbb{C}$ of p , there exists a chart $\phi_2: U_2 \subseteq Y \rightarrow V_2 \subseteq \mathbb{C}$ for $F(p)$ such that the transition function $\phi_2 \circ F \circ \phi_1^{-1}$ is holomorphic (resp. meromorphic) at $\phi_1(p)$. The map is *holomorphic* (resp. *meromorphic*) if it is holomorphic (resp. meromorphic) at every $p \in X$. A holomorphic map $F: X \rightarrow Y$ is an *isomorphism* if it is bijective and its inverse is also holomorphic. An isomorphism is an *automorphism* if $X = Y$.

For a meromorphic map $F: X \rightarrow Y$, the *order of F at p* , denoted by $\text{ord}_p F$, is the order of the transition function at $\phi_1(p)$. This does not depend on the choice of chart. For every $p \in X$ there exists charts for p and $F(p)$ such that the transition map is $z \mapsto z^m$ for some integer $m \geq 0$. This integer is the *multiplicity of F at p* , denoted as $e_p F$, and it's independent of the choice of chart. The *degree of F* is the sum

$$\deg F = \sum_{p \in F^{-1}(y)} e_p F.$$

This sum gives the same number for every $y \in Y$, so it's well defined. We say $p \in X$ is a *ramification point for F* if $\text{mult}_p F \geq 2$, and we say $F(p)$ is a *branch point for F* . The set of ramification points is discrete, and if X, Y are compact, then this set has to be finite.

More information on these concepts can be found in [3].

Theorem 1.1. *Let X be a compact Riemann surface, then there exists an integer $g(X) \geq 0$ such that, for every triangulation on X with V vertices, E edges and T triangles, we have*

$$V - E + T = 2 - 2g(X).$$

The integer $g(X)$ is called the *genus* of X .

2. RIEMANN-HURWITZ THEOREM

Given $F: X \rightarrow Y$ a holomorphic function between compact Riemann surfaces, for every $y \in Y$ we denote

$$S_y = \sum_{p \in F^{-1}(y)} (e_p F - 1).$$

Since both X and Y are compact, the set of branch points in Y is finite, and therefore the sum is well defined. We can also write

$$S_y = \sum_{p \in F^{-1}(y)} e_p F - \sum_{p \in F^{-1}(y)} 1 = \deg F - \#F^{-1}(y).$$

This is always a non-negative integer, and $S_y = 0$ if and only if y is not a branch point.

Theorem 2.1 (Riemann-Hurwitz). *Let X, Y be compact Riemann surfaces. Suppose there is a non-constant holomorphic map $F: X \rightarrow Y$. Then*

$$2g(X) - 2 = (2g(Y) - 2) \deg F + \sum_{p \in X} (e_p F - 1).$$

Proof. Consider a triangulation for Y with V vertices, E edges and T triangles. By refinement, we can assume that every branch point is in a vertex of the triangulation. This lifts as a triangulation in X with V' vertices, E' edges and T' triangles. Since every branch point is in a vertex, we have $E' = E \deg F$ and $T' = T \deg F$. for the vertices we have

$$V' = \sum_{\substack{q \in Y \\ \text{vertex}}} \#F^{-1}(q) = \sum_{\substack{q \in Y \\ \text{vertex}}} (\deg F - S_q) = V \deg F - \sum_{y \in Y} S_y = V \deg F - \sum_{p \in X} (e_p F - 1).$$

Using that $2g(X) - 2 = -V' + E' - T'$ and $2g(Y) - 2 = -V + E - T$, we get the result. $\square$

We also have an alternative form of the formula:

$$2g(X) - 2 = (2g(Y) - 2) \deg F + \sum_{y \in Y} S_y.$$

Now we will see some applications of this theorem.

3. APPLICATIONS ON THE GENUS

This theorem gives interesting results on the genus of compact Riemann surfaces.

Proposition 3.1. *Let $F: X \rightarrow Y$ a non-constant holomorphic map of compact Riemann surfaces, then $g(X) \geq g(Y)$.*

Proof. Note that $\deg F \geq 1$ and $S_y \geq 0$. Then apply Theorem 2.1. $\square$

With this, we can get some interesting bounds for the genus of certain Riemann surfaces. The following is a direct consequence of this property.

Theorem 3.2 (FLT for polynomials). *Let $f, g, h \in \mathbb{C}[t]$ be non-constant polynomials such that $f^n + g^n = h^n$. then $n \leq 2$.*

Proof. Consider the smooth projective planar curves C_n given by $x^n + y^n = z^n$. Consider the function $F: \mathbb{P}^1 \rightarrow C_n$ given by

$$F(t) = [f(t) : g(t) : h(t)].$$

This is a holomorphic map between compact Riemann surfaces, and then

$$0 = g(\mathbb{P}^1) \geq g(C_n) = \frac{(n-1)(n-2)}{2}.$$

Hence $n \leq 2$. □

Another useful tool is to consider an algebraic curve X and then take the map $\pi: X \rightarrow \mathbb{P}^1$ given by projection to the coordinate x . An example of this is the following:

Theorem 3.3. *Let E be an elliptic curve over $\mathbb{C}$ given by $y^2 = x^3 + ax + b$, with $4a^3 - 27b^2 \neq 0$. Then $g(E) = 1$.*

Proof. Consider the projection on the x coordinate $\pi: E \rightarrow \mathbb{P}^1$, then $\deg \pi = 2$ and there are 4 ramification points, which are the 3 distinct roots of $x^3 + ax + b$ and the point at infinity. For a branch point q , we have $S_q = 2 - \#\pi^{-1}(q) \geq 1$, then $\#\pi^{-1}(q) = 1$ and therefore $S_q = 1$. By Theorem 2.1 we get

$$2g(E) - 2 = -2 \cdot 2 + 4 = 0 \implies g(E) = 1.$$

□

From the classification of smooth compact orientable manifolds, which can be found in [4] (§77 and §78), we have the following.

Corollary 3.4. *Every elliptic curve over $\mathbb{C}$ is isomorphic to a complex torus.*

4. HYPERELLIPTIC RIEMANN SURFACES

Given an integer $g \geq 0$, we can construct a compact Riemann surface of genus g as follows: Consider a polynomial $h \in \mathbb{C}[x]$ of degree $2g + 1 + \epsilon$, where $\epsilon \in \{0, 1\}$, and suppose h has distinct roots. Let X be the curve defined by the equation

$$y^2 = h(x).$$

Consider the polynomial

$$k(z) = z^{2g+2} h\left(\frac{1}{z}\right).$$

This is a polynomial with distinct roots. Let Y be the curve given by the equation

$$w^2 = k(z).$$

Consider the function

$$\begin{aligned} \phi: \{(x, y) \in X: x \neq 0\} &\rightarrow \{(z, w) \in Y: z \neq 0\}, \\ (x, y) &\mapsto \left(\frac{1}{x}, \frac{y}{x^{g+1}}\right) \end{aligned}$$

Theorem 4.1. *With the above construction, let $Z = (X \sqcup Y)/\phi$, and let $\pi: Z \rightarrow \mathbb{P}^1$ be the projection on the x -coordinate. Then Z is a compact Riemann surface of genus g , and π is a holomorphic map of degree 2 where the ramification points are the roots of h , and ∞ is a ramification point if and only if $\deg h = 2g + 1$.*

Proof. There is a unique complex structure for Z where the inclusions $X \rightarrow Z$ and $Y \rightarrow Z$ are holomorphic, then Z is a Riemann surface. Note that

$$Z = \{(x, y) \in X: \|x\| \leq 1\} \cup \{(z, w) \in Y: \|z\| \leq 1\}.$$

Then Z is the union of two compact sets, and hence it is a compact Riemann surface. The projection $\pi: Z \rightarrow \mathbb{P}^1$ is holomorphic of degree 2. Then every branch point $y \in Y$ has $S_y > 0$, therefore $\#F^{-1}(y) = 1$. Note that the roots of h gives branch points of π , and if $\deg h = 2g + 1$, then ∞ together with the $2g + 1$ are branch points. Any other point $y \in Y$ satisfies $S_y = 0$, so there are $2g + 2$ branch points, each with $S_y = 1$, and by Theorem 2.1 we get

$$2g(Z) - 2 = -2 \cdot 2 + 2g + 2 = 2g - 2 \implies g(Z) = g.$$

□

5. MASON-STOTHERS THEOREM AND THE ABC CONJECTURE

The *abc conjecture* is one of the most important conjectures in number theory, mostly because of its implications, some of which can be found in [6].

Conjecture 5.1 (*abc conjecture*). *For each $\epsilon > 0$, there exists $\kappa = \kappa(\epsilon) > 0$ such that, if $a, b, c > 0$ are pairwise coprime integers with $a + b = c$, then*

$$c < \kappa \cdot \text{rad}(abc)^{1+\epsilon}.$$

Another way of stating this is: For any $\epsilon > 0$, the set of triples (a, b, c) where $a, b, c > 0$ are pairwise coprime integers such that $a + b = c$ and

$$c > \text{rad}(abc)^{1+\epsilon}$$

is finite.

One of the most famous implications of Conjecture 5.1 is Fermat's Last Theorem:

Theorem 5.2 (FLT). *Let $x, y, z > 0$ be integers such that $x^n + y^n = z^n$. Then $n \leq 2$.*

Assuming Conjecture 5.1, we can prove that there is only a finite amount of possible counterexamples, and then give a bound for these, but it is also conjectured that for $\epsilon = 1$ the constant is $\kappa = 1$. Assuming this, we get a better proof. By dividing for common factors, we can assume that x, y, z are pairwise coprime. Then

$$z^n < \text{rad}(x^n y^n z^n)^2 = \text{rad}(xyz)^2 < (xyz)^2 < z^6$$

Then $n \leq 5$. It is well known that $n \notin \{3, 4, 5\}$, so $n \leq 2$.

For the study of Riemann surfaces, we have an equivalent of Conjecture 5.1, which is the following.

Theorem 5.3 (Mason-Stothers). *Let $f, g, h \in \mathbb{C}[t]$ be non-constant, pairwise coprime polynomials such that $f + g = h$. Then*

$$\max\{\deg f, \deg g, \deg h\} \leq \deg \text{rad}(fgh) - 1.$$

Proof. We can assume $\deg f > \deg g$. Then $u = f/h: \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is a holomorphic map between compact Riemann surfaces of degree $\deg u = \max\{\deg f, \deg h\}$. Note that, if ∞ is not a root of f, g, h , then

$$f^{-1}(0) = u^{-1}(0), \quad g^{-1}(0) = u^{-1}(1), \quad h^{-1}(0) = u^{-1}(\infty).$$

In general, we get that

$$\begin{aligned} \#u^{-1}(\{0, 1, \infty\}) &\leq \#f^{-1}(0) + \#g^{-1}(0) + \#h^{-1}(0) + 1 \\ &= \deg \text{rad } f + \deg \text{rad } g + \deg \text{rad } h + 1 \\ &= \deg \text{rad}(fgh) + 1. \end{aligned}$$

Then, by Theorem 2.1 we get

$$\begin{aligned}
-2 &= -2 \deg u + \sum_{p \in \mathbb{P}^1} S_p \\
&\geq -2 \deg u + S_0 + S_1 + S_\infty \\
&= -2 \deg u + 3 \deg u - \#u^{-1}(\{0, 1, \infty\}) \\
&\geq \deg u - \deg \text{rad}(fgh) - 1.
\end{aligned}$$

From this we get the desired result. $\square$

The standard proof of Theorem 5.3 deals with derivatives of polynomials, but this is an adaptation of Silverman's proof, which can be found in [5].

With this, we can give another proof for Theorem 3.2. Let $f, g, h \in \mathbb{C}[t]$ be non-constant pairwise coprime polynomials such that $f^n + g^n = h^n$ for some integer $n > 0$. Then by Theorem 5.3

$$M := \max\{\deg f^n, \deg g^n, \deg h^n\} \leq \deg \text{rad}(f^n g^n h^n) - 1$$

We have a lower bound for M given by the arithmetic median of the set:

$$M \geq \frac{\deg f^n + \deg g^n + \deg h^n}{3} = \frac{n \deg(fgh)}{3}.$$

Also note that

$$M \leq \deg \text{rad}(f^n g^n h^n) - 1 = \deg \text{rad}(fgh) - 1 < \deg(fgh).$$

Therefore

$$\frac{n \deg(fgh)}{3} \leq M < \deg(fgh) \implies n < 3.$$

6. HURWITZ'S AUTOMORPHISM THEOREM

Theorem 6.1. *Let X be a compact Riemann surface of genus $g \geq 2$, then $\text{Aut}X$ is finite, and in particular*

$$|\text{Aut}X| \leq 84(g-1).$$

Proof. To prove that $\text{Aut}X$ is finite we need homology. proof of this can be found on [2]. We have that $\text{Aut}X$ acts faithfully on X , and this action is holomorphic. Then we can give a complex structure to $Y = X/\text{Aut}X$, so Y is a compact Riemann surface. Consider the projection $\pi: X \rightarrow Y$, and let $y \in Y$ be a branch point. Let $\pi^{-1}(y) = \{x_1, \dots, x_s\}$, then $\pi^{-1}(y) = \text{orb}x_i$ for every $i = 1, \dots, s$. This implies that $\text{stab}x_i = \text{stab}x_1 =: r$ for every $i = 1, \dots, s$, and hence $\#\pi^{-1}(y) = |\text{Aut}X|/r$. Note that $\deg \pi = |\text{Aut}X|$, therefore $S_y = |\text{Aut}X|(1 - \frac{1}{r})$. Then by Theorem 2.1

$$2g - 2 = |\text{Aut}X| \left(2g(\text{Aut}X) - 2 + \sum_{i=1}^k \left(1 - \frac{1}{r_i} \right) \right)$$

for some integers $r_i \geq 2$. Let $R = \sum_{i=1}^k (1 - \frac{1}{r_i})$, then

$$|\text{Aut}X| = \frac{2}{2g(Y) - 2 + R}(g-1).$$

It can be proven that $2/(2g(Y) - 2 + R) \leq 84$, hence we get the desired result. $\square$

A *Hurwitz surface* is a Riemann surface X such that $|\text{Aut}X| = 84(g-1)$. The Hurwitz surface of lowest genus is the *Klein quartic*, which is given by the equation

$$X : x^3y + y^3z + z^3x = 0.$$

This is a smooth planar projective curve, and it has genus 3. By Theorem 2.1, we have

$$|\mathrm{Aut} X| \leq 84(3 - 1) = 168.$$

Proposition 6.2. *The Klein quartic has 168 automorphisms. more specifically, its automorphism group is isomorphic to $\mathrm{PSL}_2(\mathbb{F}_7)$.*

This can be found in [1].

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